

Tonal Languages

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```
glm_P_sq <- glm(data = tones_complex,
                formula = complex_tonal ~ I(dist_equator^2),
                family = binomial(link = probit))
summary(glm_P_sq)

##
## Call:
## glm(formula = complex_tonal ~ I(dist_equator^2), family = binomial(link = probit),
##      data = tones_complex)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7424  -0.7169  -0.6202  -0.2209   2.1963
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.7034691   0.0811720  -8.666  < 2e-16 ***
## I(dist_equator^2) -0.0005245   0.0001208  -4.343  1.4e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 475.43  on 526  degrees of freedom
## Residual deviance: 446.95  on 525  degrees of freedom
## AIC: 450.95
##
## Number of Fisher Scoring iterations: 7
```

Manual Error Analysis

Idea: We try to manually find out why equator-close languages are atonal from other variables, and add potential variables (for example, colonization history, case of extreme language erasure, etc.) In Thailand, it was under French Indochina, but it survived? So, need to learn more about each case and find potential sociopolitical factors to explain more variation. This would come from manually studying each case and history.

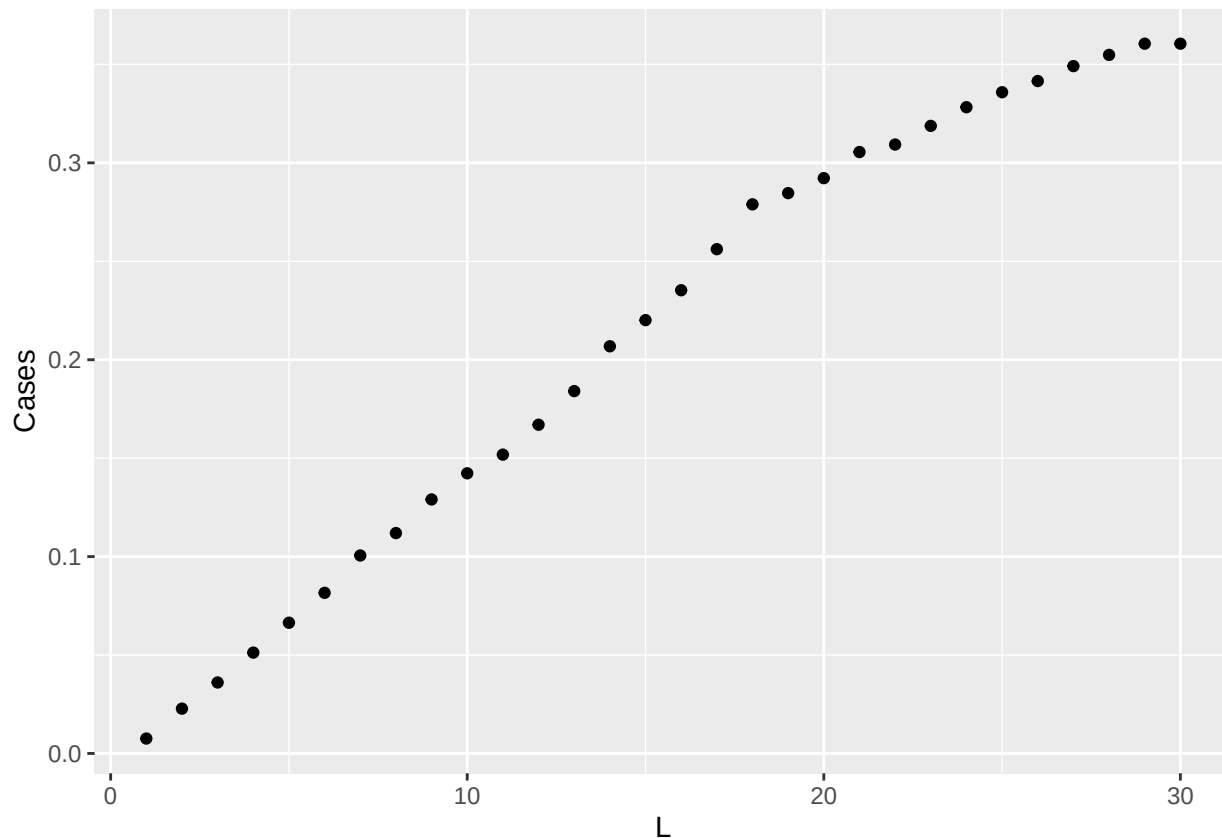
In data visualization, we found that no tonal language existed farther than 30 latitude. So, we use that as our reference point in order to try to obtain perfect prediction. However, in practice, we might want to use alternative thresholds to really emphasize the outlier status.

A no-brainer: incorporate climatically outlier zones, where equator doesn't imply humidity, if possible.

```
# Make a copy of the data
outliers <- tones_complex
```

```
table(tones_complex$complex_tonal)
```

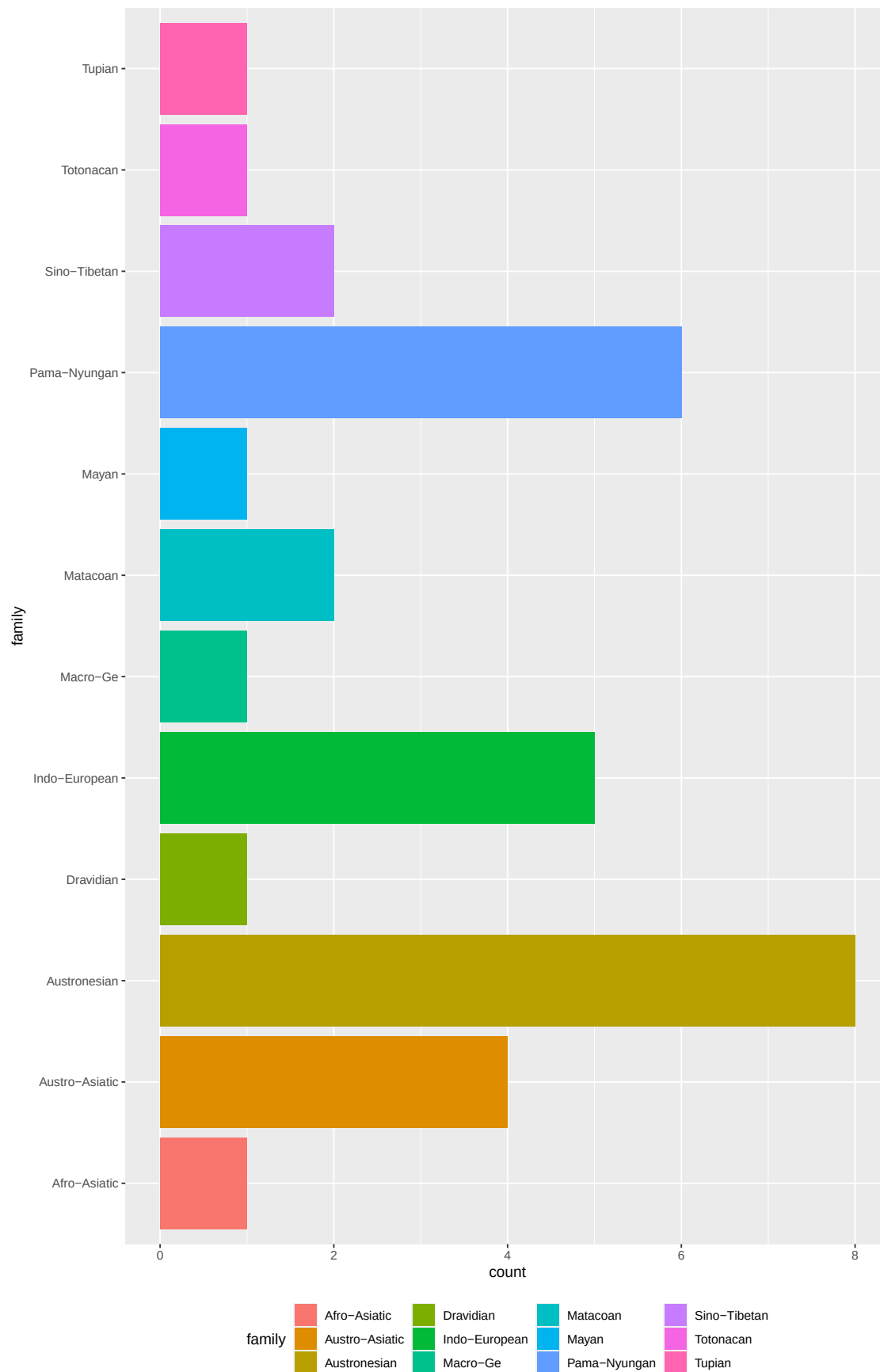
```
##
## FALSE TRUE
## 439 88
```



```
study <- outliers %>%
  filter(dist_equator < 30) %>%
  filter(dist_equator > 20) %>%
  filter(tonality_type == 1)
```

```
study %>%
  ggplot() +
  geom_histogram(aes(x = family, fill = family), stat = "count") +
  coord_flip() +
  theme(legend.position = "bottom")
```

```
## Warning in geom_histogram(aes(x = family, fill = family), stat = "count"):
## Ignoring unknown parameters: `binwidth`, `bins`, and `pad`
```



```

# Get ggplot coords
world_coordinates <- map_data("world")
# Plot
study %>%
  ggplot() +
  geom_map(data = world_coordinates, map = world_coordinates,
           aes(long, lat, map_id = region)) +
  geom_point(aes(x = longitude, y = latitude), alpha = 0.3, color = "red")

```

```

## Warning in geom_map(data = world_coordinates, map = world_coordinates,
## aes(long, : Ignoring unknown aesthetics: x and y

```

